CHAPTER 4

Nuclear Mass Stability

Contents

4.1. Patterns of Nuclear Stability 65
4.2. Neutron to Proton Ratio 67
4.3. Mass Defect 69
4.4. Binding Energy 72
4.5. Nuclear Radius 74
4.6. Semiempirical Mass Equation 75
4.7. Valley of β-Stability 77
4.8. The Missing Elements: $_{43}$Tc and $_{61}$Pm 79
4.8.1. Promethium 79
4.8.2. Technetium 80
4.9. Other Modes of Instability 82
4.10. Exercises 82
4.11. Literature 83

4.1. PATTERNS OF NUCLEAR STABILITY

There are today 255 different nuclei which have shown no evidence of radioactive decay and, hence, are said to be stable with respect to radioactive decay. Currently this means that if they have a half life it is greater than $\sim 10^{20}$ years, which is the longest half-life measured to this date. When these nuclei are compared for their constituent nucleons, we find that approximately 60% of them have both an even number of protons and an even number of neutrons (*even-even nuclei*). The remaining 40% are about equally divided between those that have an even number of protons and an odd number of neutrons (*even-odd nuclei*) and those with an odd number of protons and an even number of neutrons (*odd-even nuclei*). There are now only 4 stable nuclei known which have both an odd number of protons and odd number of neutrons (*odd-odd nuclei*); $^{2}_{1}$H, $^{6}_{3}$Li, $^{10}_{5}$B, and $^{14}_{7}$N, because recently $^{50}_{23}$Vwas found to be instable with a very long half-life. It is significant that the first stable odd-odd nuclei are abundant in the very light elements (the low abundance of $^{2}_{1}$H has a special explanation, see Ch. 12). The last nuclide is found in low isotopic abundance (0.25%) and has an extremely long half-life, 1.4×10^{17} y.

Considering this pattern for the stable nuclei, we can conclude that nuclear stability is favoured by even numbers of protons and neutrons. The validity of this statement can be confirmed further by considering for any particular element the number and types of stable isotopes; see Figure 4.1. Elements of even atomic number (i.e. even number

Radiochemistry and Nuclear Chemistry
ISBN 978-0-12-405897-2, http://dx.doi.org/10.1016/B978-0-12-405897-2.00004-5

© 2013 Elsevier Inc.
All rights reserved.

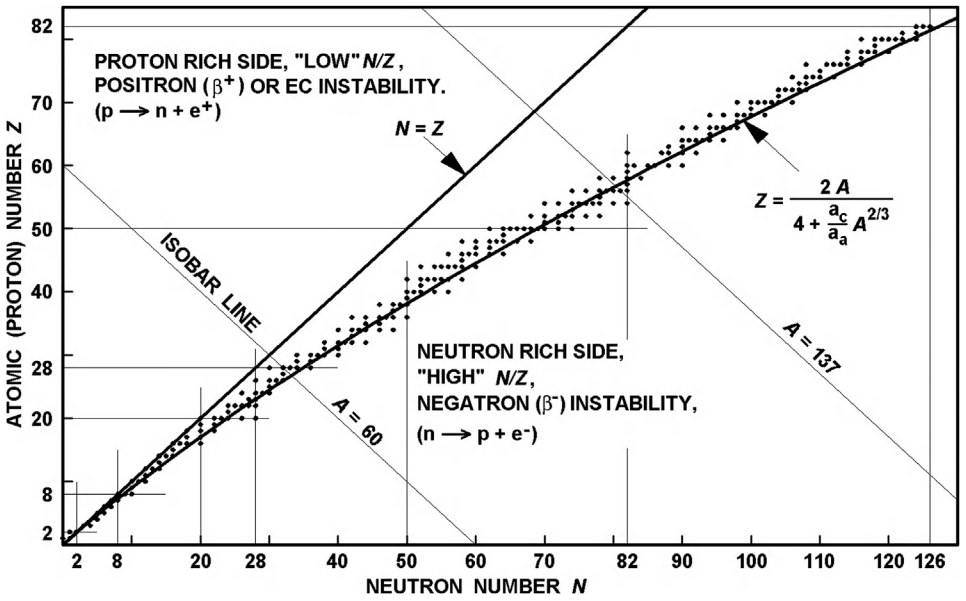


Figure 4.1 Chart of stable nuclides as a function of their proton (*Z*) and neutron (*N*) numbers. The numbers denoted 2, 8, etc., are discussed in Chapter 11.

of protons) are characterized by having a relatively sizable number of stable isotopes, usually 3 or more. For example, the element tin, atomic number 50, has 10 stable isotopes while cadmium ($Z = 48$) and tellurium ($Z = 52$) each have 8. By contrast silver ($Z = 47$) and antimony ($Z = 51$) each have only 2 stable isotopes, and rhodium ($Z = 45$), indium ($Z = 49$), and iodine ($Z = 53$) have only 1 stable isotope. Many other examples of the extra stabilization of even numbers of nucleons can be found from a detailed examination of Figure 4.1, or, easier, from nuclide charts, e.g. Appendix C. The guide lines of N and Z equal to 2, 8, 20, etc., have not been selected arbitrarily. These proton and neutron numbers represent unusually stable proton and neutron configurations, as will be discussed further in Chapter 6. The curved line through the experimental points is calculated based on the liquid drop model of the nucleus which is discussed later in this chapter.

Elements of odd Z have none, one or two stable isotopes, and their stable isotopes have an even number of neutrons, except for the 5 odd-odd nuclei mentioned above. This is in contrast to the range of stable isotopes of even Z, which includes nuclei of both even and odd N, although the former outnumber the latter. Tin ($Z = 50$), for example, has 7 stable even-even isotopes and only 3 even-odd ones.

The greater number of stable nuclei with even numbers of protons and neutrons is explained in terms of the energy stabilization gained by combination of like nucleons to form pairs, i.e. protons with protons and neutrons with neutrons, but not protons with

neutrons. If a nucleus has, for example, an even number of protons, all these protons can exist in pairs. However, if the nucleus has an odd number of protons, at least one of these protons must exist in an unpaired state. The increase in stability resulting from complete pairing in elements of even Z is responsible for their ability to accommodate a greater range of neutron numbers as illustrated for the isotopes of germanium ($_{32}$Ge, 5 stable isotopes), relative to those of gallium ($_{31}$Ga, 2 stable isotopes), and arsenic ($_{33}$As, 1 stable isotope). The same pairing stabilization holds true for neutrons so that an even-even nuclide which has all its nucleons, both neutrons and protons, paired represents a quite stable situation. In the elements in which the atomic number is even, if the neutron number is uneven, there is still some stability conferred through the proton-proton pairing. For elements of odd atomic number, unless there is stability due to an even neutron number (neutron-neutron pairing), the nuclei are radioactive with rare exceptions. We should also note that the number of stable nuclear species is approximately the same for even-odd and odd-even cases. The pairing of protons with protons and neutrons with neutrons must thus confer approximately equal degrees of stability to the nucleus. This behaviour is rather similar to the electronic structure of the atoms where also paired electrons offer a greater stability. The similarities between the nucleons in the nucleus and the electrons in the atoms is further discussed in combination with the shell model for nuclear stability, Ch. 6.

4.2. NEUTRON TO PROTON RATIO

If a graph is made (Fig. 4.1)[1] of the relation of the number of neutrons to the number of protons in the known stable nuclei, we find that in the light elements stability is achieved when the number of neutrons and protons are approximately equal ($N = Z$). However, with increasing atomic number of the element (i.e. along the Z-line), the ratio of neutrons to protons, the N/Z ratio, for nuclear stability increases from unity to about 1.52 at lead. Thus pairing of the nucleons is not a sufficient criterion for stability: a certain ratio N/Z must also exist. However, even this does not suffice for stability, because at high Z-values, a new mode of radioactive decay, α-emission, appears. Above bismuth the nuclides are all unstable to radioactive decay by α-particle emission, while some are unstable also to β-decay.

If a nucleus has an N/Z ratio too high for stability, it is said to be *neutron-rich*. It will undergo radioactive decay in such a manner that the neutron to proton ratio decreases to approach more closely the stable value. In such a case the nucleus must decrease the value of N and increase the value of Z, which can be done by conversion of a neutron to a proton. When such a conversion occurs within a nucleus, β^- (or *negatron*) *emission* is the

[1] In graphs like Fig. 4.1, Z is commonly plotted as the abscissa; we have here reversed the axes to conform with the commercially available isotope and nuclide charts.